

SOLUTIONS

illuminati - 2019

Advanced Mathematics Assignment - 2B | Conic Section

Section 1

Single Correct Answer Type

- 1.(C) Equation of tangent at vertex to parabola is

$$y - 4 = \frac{6-4}{4-3}(x-3) \Rightarrow 2x - y = 2 \quad \dots\dots\dots (1)$$

Equation of axis of parabola is $x + 2y = \lambda$ it passes through (1, 2)

$$\text{so, } x + 2y = 5 \quad \dots\dots\dots (2)$$

Solving (1) and (2) we get

$$\text{vertex } \left(\frac{9}{5}, \frac{8}{5} \right)$$

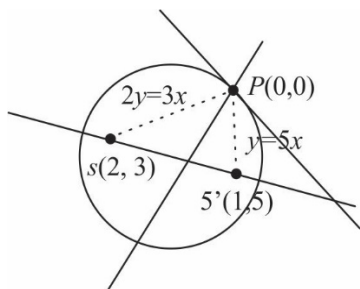
- 2.(A) Tangent and normal are bisectors of $\angle SPS'$. Now equation of SP is $y = \frac{3x}{2}$ and that of $S'P$ is $y = 5x$.

$$\text{Then their equations are } \frac{3x-2y}{\sqrt{13}} = \pm \frac{5x-y}{\sqrt{26}}$$

$$\Rightarrow \text{ lines are } (3\sqrt{2}-5)x + (1-2\sqrt{2})y = 0 \text{ and } (3\sqrt{2}+5)x - (2\sqrt{2}+1)y = 0$$

Now (2,3) and (1,5) lie on the same side of

$$(3\sqrt{2}-5)x + (1-2\sqrt{2})y = 0 \text{ which is equation of tangent.}$$



Points (2,3) and (1,5) lie on the different sides of $(3\sqrt{2}+5)x - (2\sqrt{2}+1)y = 0$, which is equation of normal.

- 3.(A) The points (4,4), (2,4) and (6,8), (0,8) are image of each other in the line $x = 3$.

$$\Rightarrow \text{ The axis of parabola is } x = 3$$

$$\Rightarrow \text{ The parabola becomes } (x-3)^2 = 4a(y+\alpha)$$

$$\Rightarrow 1 = 4a(4+\alpha) \text{ and } 9 = 4a(8+\alpha) \Rightarrow a = \frac{1}{2} \text{ and } \alpha = -\frac{7}{2}$$

$$\Rightarrow (x-3)^2 = 2\left(y - \frac{7}{2}\right) \text{ is the equation of required parabola.}$$

4.(C) Let equation of line passing through $P(1, t)$ be

$$\frac{x-1}{\cos \theta} = \frac{y-t}{\sin \theta} = r$$

$$\Rightarrow x = r \cos \theta + 1 \text{ and } y = r \sin \theta + t$$

Line meets the parabola at A and B

$$\Rightarrow (r \sin \theta + t)^2 = 4(r \cos \theta + 1)$$

$$\Rightarrow r^2 \sin^2 \theta + 2r(t \sin \theta - 2 \cos \theta) + t^2 - 4 = 0$$

$$\therefore PA \cdot PB = \left| \frac{t^2 - 4}{\sin^2 \theta} \right| = 3|t|$$

$$\Rightarrow \frac{|t^2 - 4|}{3|t|} = \sin^2 \theta \leq 1 \Rightarrow \frac{(|t|^2 - 4)^2}{9|t|^2} \leq 1$$

$$\Rightarrow 1 \leq |t| \leq 4 \Rightarrow t \in [-4, -1] \cup [1, 4]$$

5.(B) Any point on the curve

$$b^4x + 2a^2y^2 = 0 \text{ is } \left(-\frac{2a^2t^2}{b^4}, t \right).$$

Equation of QR is

$$\frac{x}{a^2} \left(-\frac{2a^2t^2}{b^4} \right) - \frac{yt}{b^2} = 1 \Rightarrow -\frac{2xt^2}{b^4} - \frac{yt}{b^2} = 1$$

$$-2xt^2 - ytb^2 = b^4 \quad \dots\dots\dots (1)$$

Now we have to prove that (1) touches a fixed parabola

From (1), we have $b^2ty = -2xt^2 - b^4$

$$\Rightarrow y = \frac{-2xt^2}{b^2t} - \frac{b^4}{b^2t} \Rightarrow y = \frac{-2t}{b^2}x - \frac{b^2}{t}$$

$$y = \left(-\frac{2t}{b^2} \right)x + \frac{2}{(-2t/b^2)} \quad \dots\dots\dots (2)$$

Clearly (2) is of the form $y = mx + \frac{2}{m}$, which always touches the parabola $y^2 = 8x$ for all real values of t .

6.(D) Let (h, k) be the focus of parabola.

The foot of perpendicular from (h, k) to $x = a$ is (a, k)

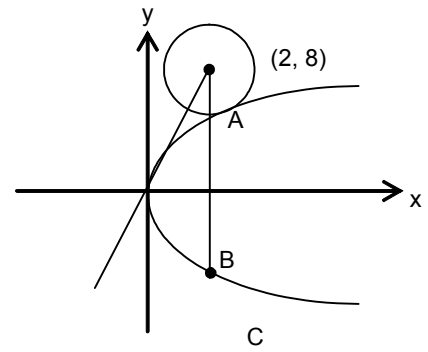
Similarly, for $y = b$ it is (h, b)

As perpendicular from the focus on any tangent meets it on the tangent at vertex Hence the slope of tangent at vertex is $\frac{k-b}{a-h}$ as tangent at vertex is parallel to directrix.

$$\Rightarrow \frac{k-b}{a-h} = m$$

Hence the locus is $y = mx = am + b$

- 7.(C) The normal will always pass through the centre $(2, 8)$ of the circle. The length of AB is zero (minimum) when the normal becomes tangent to the parabola and AB is maximum when the normal is parallel to y -axis



$$\Rightarrow \text{For the maximum value of } AB \text{ the normal becomes } x=2$$

$$\Rightarrow y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$$

$$\Rightarrow AB = 2\sqrt{2} - (-2\sqrt{2}) = 4\sqrt{2}$$

$$\Rightarrow AB \text{ lies in } (0, 4\sqrt{2}]$$

- 8.(A) Given $S(1, -1)$ and $S'(2, -1)$ are foci.

$$\text{Hence } 2ae = SS' = \sqrt{(2-1)^2 + (-1+1)^2} = 1$$

$$\Rightarrow ae = \frac{1}{2} \quad \dots\dots\dots (1)$$

Also $P_1P_2 = b^2$, where P_1, P_2 are lengths of perpendiculars from S and S' to the tangent.

$$\text{So, } \frac{5}{\sqrt{2}} \times \frac{4}{\sqrt{2}} = b^2 \text{ or } b^2 = 10 \quad \dots\dots\dots (2)$$

$$b^2 = 10 \Rightarrow a^2(1 - e^2) = 10 \text{ or } a^2 = 10 + \frac{1}{4} = \frac{41}{4} \quad (\text{Using (1)})$$

$$\text{Hence equation of the ellipse is } \frac{\left(x - \frac{3}{2}\right)^2}{\frac{41}{4}} + \frac{(y+1)^2}{10} = 1.$$

On solving the above equation with $x + y - 5 = 0$, we get

$$81x^2 - 612x + 1156 = 0 \Rightarrow x = \frac{34}{9} \Rightarrow \text{Point of contact is } \left(\frac{34}{9}, \frac{11}{9}\right).$$

Alternate Solution

Find image S'' of S' with respect to $x + y - 5 = 0$

$$\frac{h-2}{1} = \frac{k+1}{1} = \frac{-2(-4)}{2}$$

$$\Rightarrow S'' = (6, 3).$$

Let P be the point of contact.

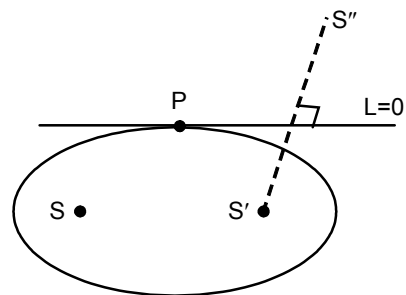
Because the line $L=0$ is tangent to the ellipse there exists a point P uniquely on the line such that

$$PS + PS' = 2a$$

$$\text{Since } PS' = PS''$$

There exists one and only one point P on $L=0$ such that $PS + PS'' = 2a$. Hence P should be the collinear with SS'' . Hence P is point of intersection of $SS''(4x - 5y = 9)$ and $x + y - 5 = 0$

$$\text{i.e. } P = \left(\frac{34}{9}, \frac{11}{9}\right)$$



- 9.(A) If the normals are concurrent at point $R(h, k)$ then the curve passing through point of intersection of lines and ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 + \lambda(px + qy - 1)(lx + my - 1) = 0 \quad \dots\dots (1)$$

Also equation of normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is at (x_1, y_1) on the ellipse is,

$$b^2 x_1 (y - y_1) - a^2 y_1 (x - x_1) = 0$$

Since this normal passes through point (h, k) , we have

$$b^2 x_1 k - a^2 h y_1 + (a^2 - b^2) x_1 y_1 = 0$$

Thus foot of normals drawn from $M(h, k)$ to the ellipse will lie on the curve, whose equation is,

$$b^2 x k - a^2 h x + (a^2 - b^2) x y = 0 \quad \dots\dots (2)$$

Since (1) and (2) are the same curve comparing the coefficients, we get $a^2 l p = b^2 m q$.

- 10.(B) Let the equation of tangent to the parabola $y^2 = 4x$ be

$$y = mx + \frac{1}{m} \quad \dots\dots (1)$$

Any point on this line can be taken as

$$P\left(\alpha, m\alpha + \frac{1}{m}\right)$$

Now, let any point on the ellipse be $B(h, k)$

$$\Rightarrow \frac{h-2}{2} = \alpha \text{ and } \frac{k+0}{2} = m\alpha + \frac{1}{m}$$

$$\Rightarrow h = 2\alpha + 2 \text{ and } k = 2\left(m\alpha + \frac{1}{m}\right)$$

Since B lies on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$

$$\Rightarrow (\alpha + 1)^2 + 4\left(m\alpha + \frac{1}{m}\right)^2 = 1$$

$$\Rightarrow \alpha^2 + 2\alpha + 1 + 4\left(m^2 \alpha^2 + \frac{1}{m^2} + 2\alpha\right) = 1$$

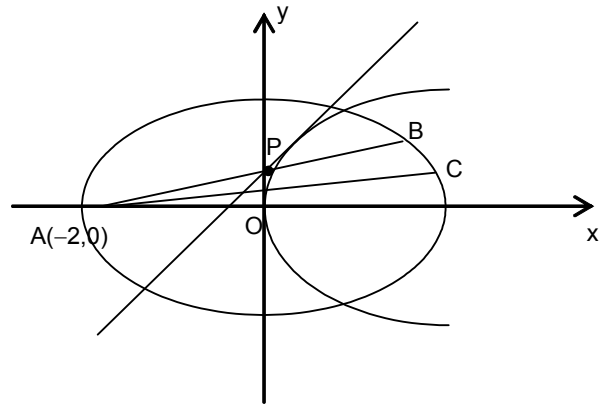
$$\Rightarrow \alpha^2 (1 + 4m^2) + 10\alpha + \frac{4}{m^2} = 0$$

Since α is real and distinct

$$\Rightarrow 100 - 4(1 + 4m^2) \frac{4}{m^2} > 0$$

$$\Rightarrow 100m^2 - 16 - 64m^2 > 0 \Rightarrow 36m^2 > 16$$

$$\Rightarrow |m| > \frac{2}{3} \Rightarrow m \in \left(-\infty, -\frac{2}{3}\right) \cup \left(\frac{2}{3}, \infty\right)$$



- 11.(C) Let the perpendicular lines be taken as the coordinate axes. Let (h, k) be the centre of the ellipse with $2a$ and $2b$ as its major and minor axes. Since the coordinate axes are touching the ellipse, the origin lies on the director circle $(x-h)^2 + (y-k)^2 = a^2 + b^2$ of the ellipse. Hence $h^2 + k^2 = a^2 + b^2$ so that locus of (h, k) is $x^2 + y^2 = a^2 + b^2$ which is a circle.

- 12.(C) The ellipse $\frac{(x-y+7)^2}{6} + \frac{(x+y-5)^2}{4} = 1$ has major axis $x+y-5=0$ and minor axis $x-y+7=0$

Now directrix will be parallel to minor axis of ellipse & will be at a distance of $\frac{a}{e}$ from minor axis.

$$\text{Now } a = \sqrt{6} \quad \& \quad e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{4}{6}} = \sqrt{\frac{2}{6}} = \frac{1}{\sqrt{3}}$$

Equation of directrix is $x-y+C=0$

$$\text{Distance between minor axis and directrix} = \frac{a}{e} = \frac{\sqrt{6} \times \sqrt{3}}{1} = \left| \frac{C-7}{\sqrt{2}} \right|$$

$$C = 7 \pm \sqrt{2} \cdot 3\sqrt{2} = 7 \pm 6$$

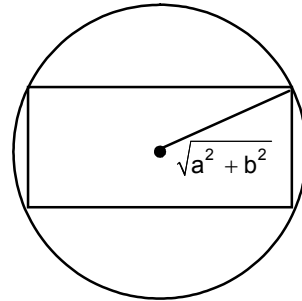
Hence directrix is $x-y+13=0$ and $x-y+1=0$.

- 13.(C) Since mutually perpendicular tangents can be drawn from vertices of rectangle. So all the vertices of rectangle should lie on director circle $x^2 + y^2 = a^2 + b^2$.

Let breadth $= 2l$ and length $= 4l$ then

$$l^2 + (2l)^2 = a^2 + b^2 \Rightarrow l^2 = \frac{a^2 + b^2}{5}$$

$$\text{Area} = 4l \cdot 2l = 8 \cdot \frac{a^2 + b^2}{5}$$



- 14.(B) Let $AB = l_1$.

$$\text{Equation of } AB \frac{x-1}{\cos \theta} = \frac{y-0}{\sin \theta} = r$$

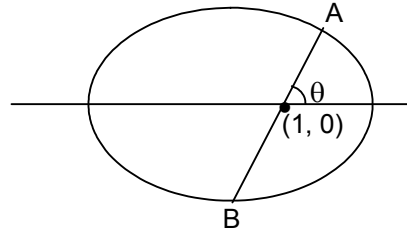
$$\Rightarrow \frac{(1+r \cos \theta)^2}{4} + \frac{(r \sin \theta)^2}{3} = 1$$

$$\Rightarrow r^2(3 + \sin^2 \theta) + 6r \cos \theta - 9 = 0$$

$$r_1 - r_2 = -\frac{6 \cos \theta}{3 + \sin^2 \theta}, r_1 r_2 = \frac{9}{3 + \sin^2 \theta}$$

$$l_1 = |r_1| + |r_2| = \sqrt{(r_1 - r_2)^2 + 4r_1 r_2} = \frac{12}{3 + \sin^2 \theta}$$

$$l_2 = \frac{12}{3 + \sin^2 \left(\frac{\pi}{2} + \theta \right)} = \frac{12}{3 + \cos^2 \theta} \Rightarrow \frac{1}{l_1} + \frac{1}{l_2} = \frac{7}{12}$$



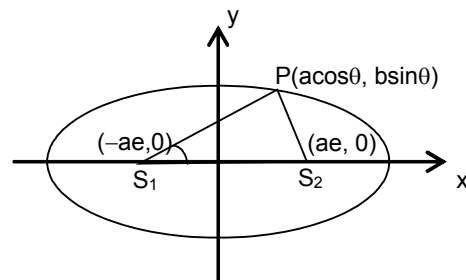
- 15.(B) Let any point P on the ellipse be $P(a \cos \theta, b \sin \theta)$

If $\angle S_1 P S_2 = \frac{\pi}{2}$, then P lies on the circle having

$S_1 S_2$ as its diameter.

$\Rightarrow$ Equation of the circle drawn on $S_1 S_2$ as diameter is $x^2 + y^2 = a^2 e^2 = a^2 - b^2$.

Since the point P should not lie on the ellipse.



$\Rightarrow$ There should not be any point of intersection of

$$x^2 + y^2 = a^2 - b^2 \text{ and } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow a^2 - b^2 < b^2 \Rightarrow 2b^2 > a^2 \Rightarrow \frac{b^2}{a^2} > \frac{1}{2}$$

$$\Rightarrow e^2 = 1 - \frac{b^2}{a^2} \Rightarrow e^2 < \frac{1}{2} \Rightarrow e \in \left(0, \frac{1}{\sqrt{2}}\right)$$

16.(D) Let $P(a \sec \theta, b \tan \theta)$ be the centre of the circle.

Let $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) are the points of intersection of circle and the rectangular hyperbola $xy = c^2$.

$$\text{So } \frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{a}{2} \sec \theta$$

$$\text{and } \frac{y_1 + y_2 + y_3 + y_4}{4} = \frac{b}{2} \tan \theta \quad [\text{Using property}]$$

Let (h, k) be the centroid of triangle formed by ordered pairs $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

$$\Rightarrow 3h = x_1 + x_2 + x_3 = 2a \sec \theta - x_4$$

$$3k = y_1 + y_2 + y_3 = 2b \tan \theta - y_4$$

$$\Rightarrow x_4 = 2a \sec \theta - 3h$$

$$y_4 = 2b \tan \theta - 3k$$

Since (x_4, y_4) also lies on the $xy = c^2$

$$\Rightarrow (2a \sec \theta - 3h)(2b \tan \theta - 3k) = c^2$$

$$\Rightarrow \left(h - \frac{2a}{3} \sec \theta\right) \left(k - \frac{2b}{3} \tan \theta\right) = \left(\frac{c}{3}\right)^2$$

$$\text{Hence locus is } \left(x - \frac{2a}{3} \sec \theta\right) \left(y - \frac{2b}{3} \tan \theta\right) = \left(\frac{c}{3}\right)^2$$

Which is a rectangular hyperbola with centre $\left(\frac{2a}{3} \sec \theta, \frac{2b}{3} \tan \theta\right)$ which lies on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \left(\frac{2}{3}\right)^2.$$

17.(D) Let the axes be two perpendicular tangents (h, k) be the centre and two foci be $(x_1, y_1), (x_2, y_2)$

$$\Rightarrow x_1 + x_2 = 2h \quad \dots(1)$$

$$\text{and } y_1 + y_2 = 2k \quad \dots(2)$$

As we know the product of length of perpendicular from the foci to any tangent is a constant.

$$x_1 x_2 = y_1 y_2 \quad \dots(3)$$

From equation (1) and (2)

$$x_2 = (2h - x_1)$$

$$y_2 = (2k - y_1)$$

Putting the values in equation (3), we get

$$x_1 2h - x_1^2 = 2k y_1 - y_1^2$$

$$\Rightarrow \text{Locus is } x^2 - y^2 = 2hx - 2ky$$

$$\text{or } (x-h)^2 - (y-k)^2 = h^2 - k^2$$

which is obviously the rectangular hyperbola having (h, k) as its centre.

18.(C) If the length of transverse axes are $2a$ and $2b$.

$s_1s_2s'_1s'_2$ is a parallelogram so

$$s'_1s_2 - s_1s_2 = s'_2s_1 - s_1s_2 = s_1s'_2 - s'_1s'_2 = s'_1s_2 - s'_1s'_2$$

$$\therefore a = b$$

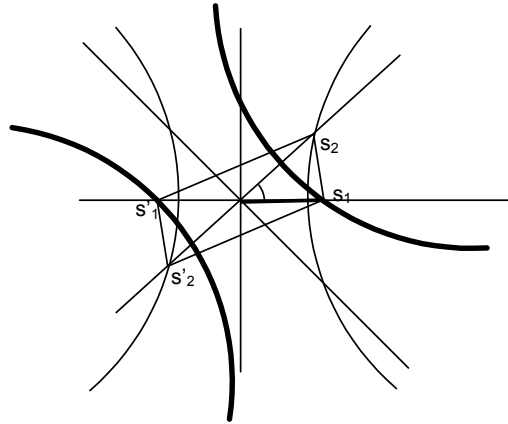
$$s'_1s_2 - s_1s_2 = 2a$$

$$s_1s_2^2 = a^2e^2 + a^2e^2 - 2a^2e^2 \cos \theta$$

$$s'_1s_2^2 = a^2e^2 + a^2e^2 - 2a^2e^2 \cos(\pi - \theta)$$

$$\Rightarrow 2a = 2a(\sqrt{1 + \cos \theta} - \sqrt{1 - \cos \theta})$$

$$\text{Solving we get } \cos^2 \theta = \frac{3}{4} \Rightarrow \theta = \frac{\pi}{6}$$



19.(D) Let hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, and tangents with slope m_1 and m_2 are $y = m_1x + \sqrt{a^2m_1^2 - b^2}$ and

$y = m_2x + \sqrt{a^2m_2^2 - b^2}$ respectively, their point of intersection with axes are $\left(\frac{-\sqrt{a^2m_1^2 - b^2}}{m_1}, 0\right)$,

$\left(0, \sqrt{a^2m_1^2 - b^2}\right)$ and $\left(\frac{-\sqrt{a^2m_2^2 - b^2}}{m_2}, 0\right)$, $\left(0, \sqrt{a^2m_2^2 - b^2}\right)$ as point are concyclic.

$$\Rightarrow \left| \sqrt{a^2m_1^2 - b^2} \sqrt{a^2m_2^2 - b^2} \right| = \left| \frac{-\sqrt{a^2m_1^2 - b^2}}{m_1} \frac{-\sqrt{a^2m_2^2 - b^2}}{m_2} \right| \Rightarrow m_1m_2 = \pm 1$$

$m_1m_2 = -1$ is rejected as points will not be concyclic.

Now, let point of intersection of tangents be (h, k)

$$\Rightarrow (k - mh)^2 = a^2m^2 - b^2 \Rightarrow (h^2 - a^2)m^2 - 2hkm + k^2 + b^2 = 0$$

$$\Rightarrow m_1m_2 = \frac{k^2 + b^2}{h^2 - a^2} = 1 \Rightarrow h^2 - k^2 = a^2 + b^2$$

locus is $x^2 - y^2 = a^2 + b^2$

20.(C) Equation of hyperbola when coordinate axes are asymptotes is $xy = \pi$

Now the equation when axes are along coordinates axis is $x^2 - y^2 = 2\pi$

$$\text{Hence length of the latus rectum} = \frac{2(2\pi)}{\sqrt{2\pi}} = 2\sqrt{2\pi}$$

21.(A) Let the rectangular hyperbola is $xy = c^2$ and $S \equiv \left(cp, \frac{c}{p}\right)$ and any point on it is $\left(ct, \frac{c}{t}\right)$.

$$\text{Equation of normal at } \left(ct, \frac{c}{t}\right) \equiv xt^3 - yt - ct^4 + c = 0$$

This normal passes through S , then

$$cpt^3 - \frac{c}{p}t - ct^4 + c = 0 \Rightarrow c(p-t)\left(t^3 \frac{1}{p}\right) = 0$$

$$\Rightarrow t^3 + \frac{1}{p} = 0 \quad (\text{as } p \neq t)$$

$$\Rightarrow t_1 + t_2 + t_3 = 0, t_1t_2 + t_2t_3 + t_3t_1 = 0 \quad \text{and} \quad t_1t_2t_3 = -\frac{1}{p} \quad \text{and centroid of the triangle}$$

$$P\left(ct_1, \frac{c}{t_1}\right), Q\left(t_2, \frac{c}{t_2}\right) \text{ and } R\left(ct_3, \frac{c}{t_3}\right) \text{ is } \left(\frac{c(t_1 + t_2 + t_3)}{3}, \frac{c\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3}\right)}{3}\right)$$

$$= \left(0, \frac{c(t_1t_2 + t_2t_3 + t_3t_1)}{3t_1t_2t_3}\right) = (0, 0) = \text{centre of hyperbola.}$$

22.(B) Equation of the tangent to the parabola $y^2 = 4ax$ at $P(x_1, y_1)$ is

$$yy_1 = 2a(x + x_1) \quad \dots\dots\dots (1)$$

Let (h, k) be any point on (1)

Equation of the chord of contact of (h, k) with respect to the given hyperbola is

$$\frac{xh}{a^2} - \frac{yk}{b^2} = 1 \quad \dots\dots\dots (2)$$

Now, (h, k) lies on (1) and (x_2, y_2) lies on (2)

$$ky_1 = 2a(h + x_1) \quad \text{and} \quad \frac{hx_2}{a^2} - \frac{ky_2}{b^2} = 1$$

comparing these equations, we get

$$\frac{2a}{x_2/a^2} = \frac{2ax_1}{-1} \Rightarrow x_1x_2 = -a^2$$

23.(C) First of all we will find locus of a point which is such that chord of contact of that point with respect to hyperbola $xy = c^2$ is normal to it.

Let that point be (h, k) chord of contact is

$$hy + kx = 2c^2 \quad \dots\dots\dots (\text{A})$$

Equation of normal at point $(ct, c/t)$

$$t^3x - ty + c - ct^4 = 0 \quad \dots\dots\dots (\text{B})$$

Since chord of contact is normal to hyperbola $xy = c^2$, Hence **(A)** and **(B)** are identical equation

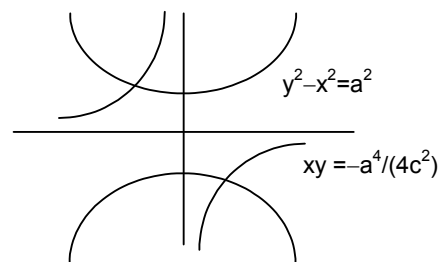
$$\Rightarrow \frac{t^3}{k} = \frac{-t}{h} = \frac{t^4 - 1}{2c} \Rightarrow t^2 = \frac{-k}{h}$$

$$\Rightarrow \text{locus is } (x^2 - y^2)^2 + 4c^2xy = 0 \quad \dots\dots\dots (1)$$

Now there will be clearly two points of intersection between the curve (1) and

$$y^2 - x^2 = a^2$$

Because by solving (1) with $y^2 - x^2 = a^2$ (1) becomes $xy = \frac{-a^4}{4c^2}$.

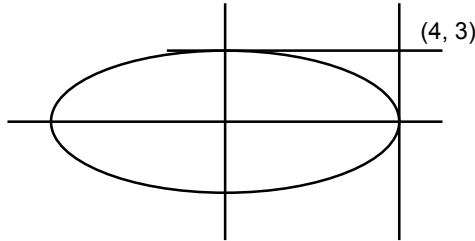


24.(B) Here $a^2 + b^2 = 1 \Rightarrow a^2 + a^2(1 - e^2) = 1 \Rightarrow 2 - e^2 = \frac{1}{a^2} \Rightarrow e^2 = \frac{2a^2 - 1}{a^2}$

$\Rightarrow$ Extremity of latus rectum is $\left(\pm\sqrt{2a^2 - 1}, \pm\frac{1 - a^2}{a} \right)$ on eliminating a , we get

$\Rightarrow 2y^2(1 + x^2) = (1 - x^2)^2$

25.(A) Point is (4, 3) from the figure centre (0, 0) is the required point.



26.(C) Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Any point on the directrix is $P \left(\frac{a}{e}, k \right)$

Chord of contact of P w.r.t the ellipse is

$$\frac{a}{e} \frac{x}{a^2} + \frac{ky}{b^2} = 1 \quad \dots\dots (1)$$

chord of contact of P w.r.t the auxiliary circle is

$$\frac{a}{e}x + ky = a^2 \quad \dots\dots (2)$$

(1) and (2) intersect at $(ae, 0)$

27.(B) Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and let $A \equiv (a \cos \theta, b \sin \theta)$

Equation of AC will be, $y = \frac{b}{a} \tan \theta$

Solving with $a = \frac{a}{e}$ we get $P \equiv \left(\frac{a}{e}, \frac{b}{e} \tan \theta \right)$.

Slope of tangent at $A = -\frac{b}{a \tan \theta}$.

$$\text{Slope of } PS = \frac{\frac{b}{e} \tan \theta}{\frac{a}{e} - ae} = \frac{b \tan \theta}{a(1 - e^2)} = \frac{a}{b} \tan \theta.$$

So $\alpha = \frac{\pi}{2}$.

28.(C) A cyclic parallelogram will be a rectangle or square.

So, $\angle QPR = 90^\circ$

$\Rightarrow P$ lies on director circle of ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$

i.e. $x^2 + y^2 = 25$ is director circle of $\frac{x^2}{16} + \frac{y^2}{b^2} = 1 \Rightarrow 16 + b^2 = 25$

$b^2 = 9 \Rightarrow a^2(1 - e^2) = 9 \Rightarrow 1 - e^2 = \frac{9}{16} \Rightarrow e^2 = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$

29.(D) Since sides of square are tangent and perpendicular to each other so vertices lie on director circle

$\Rightarrow x^2 + y^2 = (a^2 - 7) + (13 - 5a) = a^2$ ($\sqrt{2}$ is side of square)

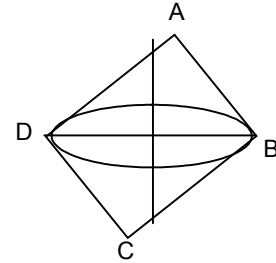
$\Rightarrow (a^2 - 7) + (13 - 5a) = a^2 \Rightarrow a = \frac{6}{5}$

but for an ellipse to exist $a^2 - 7 > 0$ and $13 - 5a > 0$

$\Rightarrow a \in (-\infty, \sqrt{7}) \cup (13/5, \sqrt{7})$

$a \neq \frac{6}{5}$

Hence no such 'a' exists.



30.(C) Major axis of hyperbola bisects the asymptote

$\Rightarrow$ Equation of other asymptote $x = 2y$

Equation of hyperbola $(y - 2x)(x - 2y) + k = 0$ it passes through (3, 4)

$\Rightarrow$ Required equation $2x^2 + 2y^2 - 5xy + 10 = 0$

31.(B) The curve is hyperbola and $e = \frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$.

32.(A) The points are such that one of the points is the orthocentre of the triangle formed by other three points. When the vertices of a triangle lie on a rectangular hyperbola the orthocentre also lies on the same hyperbola.

33.(D) We know hyperbola $xy = c^2$ and $x^2 - y^2 = a^2$

always intersects each other at right angle.

So, triangle is right angle triangle.

Circumcentre lies on y-axis.

34.(A) Clearly $a + ay - 5 = 0$ will always pass through the focus of $y^2 = 20x$ i.e. (5, 0).

Thus the drawn circle will always touch the directrix of the parabola i.e. the line $x + 5 = 0$

35.(A) Let the focus be F . The parabolas are open down and open right respectively. Let the parabolas intersect at points P and Q . From P draw perpendiculars on the x -axis and y -axis at A and B respectively, then $PA = PF = PB$

$\Rightarrow P$ lies on the line $y = -x$

Similarly Q lies on the line $y = -x \Rightarrow$ Slope of $PQ = -1$.

36.(A) The point (4, 5) lies on the given line $7x - 3y - 13 = 0$

$\Rightarrow$ The locus of the point equidistant from the given point and the line is a line perpendicular to $7x - 3y - 13 = 0$ at (4, 5).

37.(D) Here $\frac{1}{t_1} = \tan\left(\frac{\pi}{4} + \theta\right)$ and $\frac{1}{t_2} = \tan\left(\frac{\pi}{4} - \theta\right)$

So, $t_1 t_2 = 1 \Rightarrow$ the x -coordinate of $P = at_1 t_2 = a$

- 38.(B)** Let the variable parabola be $(y - \beta)^2 = 4b(x - \beta)$ where (α, β) be the vertex of parabola. Let point of intersection of the parabola $y^2 = 4ax$ and parabola $(y - \beta)^2 = 4b(x - \alpha)$ be (x_1, y_1) .

Since parabola cuts orthogonally

$$\Rightarrow m_1 m_2 = -1 \quad (\text{where } m_1 \text{ and } m_2 \text{ be slope of the tangents to parabolas } y^2 = 4ax \text{ and } (y - \beta)^2 = 4b(x - \alpha) \text{ at } (x_1, y_1) \text{ respectively}) \text{ or } m_1 = \frac{2a}{y_1} \text{ and } m_2 = \frac{2b}{y_1 - \beta}.$$

From $m_1 m_2 = -1$, we get

$$y_1^2 - \beta y_1 + 4ab = 0$$

since only one point possible where these parabolas intersect orthogonally

$$\Rightarrow \beta^2 - 4 \times 4ab = 0 \Rightarrow y^2 = 16ab.$$

- 39.(B)** For two distinct tangents on different branches the point should lie on the line $y = 2$ and between A and B (where A and B are the points on the asymptotes).

Equation of asymptotes are $4x = \pm 3y$

Solving with $y = 2$

$$x = \pm \frac{3}{2} \quad \therefore -\frac{3}{2} < \alpha < \frac{3}{2}$$

Section 2

Link Comprehension Type

40-43

- 40.(C)** $P(x, y, z)$ lies on line through $(0, 0, 2)$ making 60° with z -axis

$$\text{i.e. } \frac{(xi + yj + (z-2)k) \cdot \hat{k}}{\sqrt{x^2 + y^2 + (z-2)^2}} = \cos 60^\circ \Rightarrow x^2 + y^2 - 3(z-2)^2 = 0$$

- 41.(A)** The plane passes through $(0, 0, 2)$ and its inclination with z -axis is θ , where $\cos \theta = \frac{1}{\sqrt{3}}$

i.e. $\theta < 60^\circ \Rightarrow$ it represent a pair of straight lines.

- 42.(B)** The plane is parallel to z -axis $\Rightarrow$ locus is a hyperbola

- 43.(D)** Plane does not pass through $(0, 0, 2)$ and it is neither parallel to axis or generator.

$\Rightarrow$ locus is ellipse

44-48

- 44.(B)** Taking point $(r \cos \theta, r \sin \theta)$, we get

$$r^2 = \frac{3 + \sqrt{2}}{3 - \sqrt{2} \sin \left(2\theta + \frac{\pi}{4} \right)}$$

$$r_{\max} = \sqrt{\frac{3 + \sqrt{2}}{3 - \sqrt{2}}}, r_{\min} = 1$$

$$\text{max exist when } 2\theta + \frac{\pi}{4} = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{8}$$

45.(B) Length of semi minor axis = 1 unit $\Rightarrow$ tangent is at the end of minor axis.

so its equation can be $x \sin \frac{\pi}{8} - y \cos \frac{\pi}{8} = \pm 1$

46.(C) Product of length of perpendiculars from foci = b^2 (b is length of semi minor axis) = 1.

47.(D) Required equation is $\frac{x^2}{\frac{3+\sqrt{2}}{3-\sqrt{2}}} + y^2 = 1$

48.(A) Centre of circle be origin and its radius is the length of semi minor axis.

Hence area = π

Section 3

Multiple Correct Answer Type

49.(ACD)

50.(AB)

51.(AD)

52.(ABCD)

$$px^2 + 2qxy + ry^2 + 2sx + 2ty + u = 0 \quad \dots\dots (1)$$

will represent a pair of straight lines if

$$\begin{vmatrix} p & q & s \\ q & r & t \\ s & t & u \end{vmatrix} = 0$$

$$\therefore (1) \text{ will represent a conic if } \begin{vmatrix} p & q & s \\ q & r & t \\ s & t & u \end{vmatrix} \neq 0 \quad \dots\dots (2)$$

Further, if $D = q^2 - pr$, then (1) will represent an ellipse, parabola or hyperbola if $D <, =$ or > 0 respectively and we know that

$$\text{A.M.} > \text{G.M.} > \text{H.M.} \quad \dots\dots (3) \quad [\because p \neq q \neq r]$$

(A) If p, q, r are in H.P., then $q = \frac{2pr}{p+r} = \text{H.M. of } p, r$ using (3),

$$\sqrt{pr} > q \Rightarrow q^2 - pr < 0 \Rightarrow D < 0$$

$\therefore$ (1) represents an ellipse if p, q, r are in H.P.

(B) If p, q, r are in G.P., then $q = \text{G.M. of } p, r$

$$\Rightarrow q = \sqrt{pr} \Rightarrow q^2 - pr = 0 \Rightarrow D = 0$$

$\therefore$ (1) represents a parabola if p, q, r are in G.P.

(C) If p, q, r are in A.P., then $q = \text{A.M. of } p, r$

$$\text{Using (3), } q > \sqrt{pr} \Rightarrow q^2 - pr > 0 \Rightarrow D > 0$$

$\therefore$ (1) represents a hyperbola if p, q, r are in A.P.

(D) Obviously, above (A), (B) and (C) are true only if (2) holds.

$$53.(\text{ABC}) \quad 2 + e^{i\theta} = \frac{3}{\lambda x + i\mu y} \Rightarrow 2 + \cos \theta + i \sin \theta = \frac{3}{\lambda x + i\mu y}$$

$$\Rightarrow \cos \theta + i \sin \theta = \frac{(3 - 2\lambda x) - 2i\mu y}{\lambda x + i\mu y}$$

On taking modulus to both sides,

$$\Rightarrow 1 = \frac{\sqrt{(3-2\lambda x)^2 + (-2\mu y)^2}}{\sqrt{(\lambda x)^2 + (\mu y)^2}}$$

$$\Rightarrow \lambda^2 x^2 + \mu^2 y^2 = 9 - 12\lambda x + 4\lambda^2 x^2 + 4\mu^2 y^2$$

$$\Rightarrow \lambda^2 x^2 + \mu^2 y^2 - 4\lambda x + 3 = 0 \quad \dots\dots\dots (1)$$

This is the locus of $P(x, y)$

(A) If $\lambda = 1, \mu = 2$ then (1) becomes $x^2 + 4y^2 - 4x + 3 = 0$

$$\Rightarrow (x-2)^2 + 4y^2 = 1 \Rightarrow \frac{(x-2)^2}{1^2} + \frac{y^2}{1/4} = 1,$$

Which is an ellipse

(B) If $\mu = 0, \lambda = 1$ then (1) becomes $x^2 - 4x + 3 = 0$

$\Rightarrow x = 1, 3$; which represents a pair of straight lines.

(C) If $\lambda = \mu = 1$ then (1) becomes $x^2 + y^2 - 4x + 3 = 0$

$\Rightarrow (x-2)^2 + y^2 = 1$, which is a circle.

(D) $\mu = 1, \lambda = -1$ then (1) becomes $x^2 + y^2 + 4x + 3 = 0$

$\Rightarrow (x+2)^2 + y^2 = 1$, which is a circle and not hyperbola

54.(AD) Equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Equation of the parabola with focus $S(ae, 0)$ and directrix $x + ae = 0$ is $y^2 = 4aex$

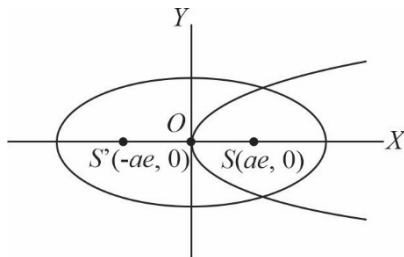
Now length of L.R. of the ellipse is $\frac{2b^2}{a}$ and that of the parabola is $4ae$.

For the two latus rectum to be equal,

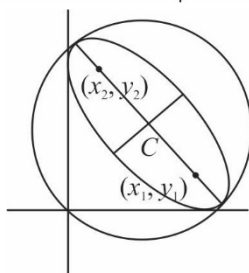
We get $\frac{2b^2}{a} = 4ae$

$$\Rightarrow \frac{2a^2(1-e^2)}{a} = 4ae \Rightarrow 1-e^2 = 2e \Rightarrow e^2 + 2e - 1 = 0 \quad \dots\dots\dots (i)$$

$$\Rightarrow e = \sqrt{2} - 1$$



55.(AC)



We know that product of length of perpendiculars from foci upon any tangents to ellipse is b^2 .

Hence from the diagram, x_1 and x_2 are length of perpendiculars from foci upon tangent, i.e. y-axis of the given ellipse, hence $x_1 \cdot x_2 = b^2$ similarly $y_1 \cdot y_2 = b^2$

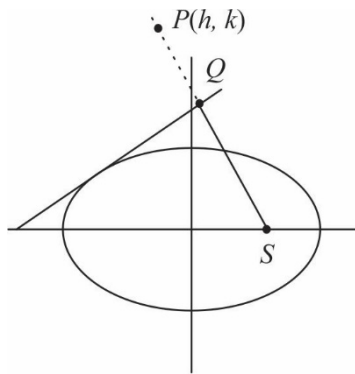
Again, we know that director circle is the locus of point of intersection of two perpendicular tangents so director circle must pass through (0, 0) and centre is mid point of (x_1, y_1) and (x_2, y_2)

$$\text{i.e. } \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{So } r = \sqrt{\left(\frac{x_1 + x_2}{2} \right)^2 + \left(\frac{y_1 + y_2}{2} \right)^2} = \frac{1}{2} \sqrt{(x_1 + x_2)^2 + (y_1 + y_2)^2}$$

56.(AB) Here $a = 5, b = 3$

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{25}} \Rightarrow e = \frac{4}{5}$$



$\Rightarrow$ focus is $(\pm 4, 0)$

Now let $P(h, k)$ be the image

So PS cuts a tangent at a point which lies on the auxiliary circle of the ellipse.

But equation of auxiliary circle is $x^2 + y^2 = 25$

and mid point of PS i.e. Q is $\left(\frac{h \pm 4}{2}, \frac{k + 0}{2} \right)$ which lies on auxiliary circle so $\left(\frac{h \pm 4}{2} \right)^2 + \left(\frac{k}{2} \right)^2 = 25$

$\Rightarrow$ locus is $(x \pm 4)^2 + y^2 = 100$

Which represent circles whose centre at $(\pm 4, 0)$ with radius 10.

57.(AB) Let the ellipse E be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

The line $y = mx \pm \sqrt{a^2 m^2 + b^2}$ touches the ellipse for all m .

Hence, it is identical with

$$y = -\frac{2px}{\sqrt{1-p^2}} + \frac{1}{\sqrt{1-p^2}}$$

$$\text{Hence } m = \frac{-2p}{\sqrt{1-p^2}} \text{ and } a^2 m^2 + b^2 = \frac{1}{1-p^2}$$

$$\Rightarrow a^2 \left(\frac{4p^2}{1-p^2} \right) + b^2 = \frac{1}{1-p^2}$$

$$\Rightarrow p^2(4a^2 - b^2) + b^2 - 1 = 0$$

This equation is true for all real p if

$$b^2 = 1 \text{ and } a^2 = \frac{1}{4}$$

Therefore, the equation of ellipse is $\frac{x^2}{1/4} + \frac{y^2}{1} = 1$

If e is its eccentricity, then

$$\frac{1}{4} = 1 - e^2 \Rightarrow e = \frac{\sqrt{3}}{2}$$

$$\text{foci}(0, \pm be) = \left(0, \pm \frac{\sqrt{3}}{2}\right)$$

and equation of director circle is $x^2 + y^2 = 1 + \frac{1}{4} = \frac{5}{4}$ whose radius is $\frac{\sqrt{5}}{2}$.

58.(ACD) $\lambda x - y + 2(1 + \lambda) = 0$

$$\Rightarrow \lambda(x + 2) - (y - 2) = 0$$

This line pass through $(-2, 2)$

$$\mu x - y + 2(1 - \mu) = 0$$

$$\Rightarrow \mu(x - 2) - (y - 2) = 0$$

This line pass through $(2, 2)$

Clearly $(2, 2)$ and $(-2, 2)$ are the foci of the ellipse. So $2ae = 4$

The circle $x^2 + y^2 - 4y - 5 = 0$

$$\Rightarrow x^2 + (y - 2)^2 = 9$$

Represents auxiliary circle.

$$\text{Thus } a^2 = 9 \Rightarrow e = 2/3 \text{ and } b^2 = 5$$

59.(ABD)

Solving both equation, we have

$$\frac{x^2}{a^2} + \frac{1 - (x - 1)^2}{b^2} = 1$$

$$\Rightarrow (b^2 - a^2)x^2 + 2a^2x - a^2b^2 = 0 \quad \dots\dots\dots (1)$$

For least area circle must touch the ellipse

$\Rightarrow$ Discriminant of equation (1) is zero

$$\Rightarrow 4a^4 + 4a^2b^2(b^2 - a^2) = 0 \Rightarrow a^2 + b^2(b^2 - a^2) = 0$$

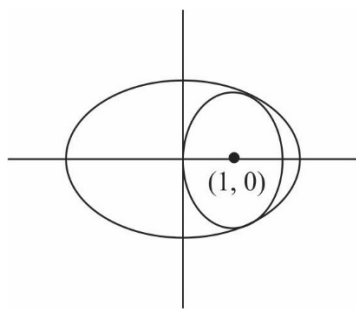
$$a^2 + b^2(-a^2e^2) = 0$$

$$\Rightarrow 1 - b^2e^2 = 0 \Rightarrow b = \frac{1}{e}$$

$$\text{Also } a^2 = \frac{b^2}{1 - e^2} = \frac{1}{e^2(1 - e^2)} \Rightarrow a = \frac{1}{e\sqrt{1 - e^2}}$$

Let S be the area of the ellipse

So, $S = \pi ab = \frac{\pi}{e^2 \sqrt{1-e^2}}$



$$= \frac{\pi}{\sqrt{e^4 - e^6}}$$

Area is minimum if $f(e) = e^4 - e^6$ is maximum so for maxima or minima $f'(e) = 4e^3 - 6e^5 = 0$

or $e = \sqrt{\frac{2}{3}}$ (which is point of maxima for $f(e)$)

$\Rightarrow$ S is least when $e = \sqrt{\frac{2}{3}} \Rightarrow$ ellipse is $2x^2 + 6y^2 = 9$

Equation of auxiliary circle of the ellipse is $x^2 + y^2 = 4.5$

Length of L.R of ellipse is $\frac{2b^2}{a} = \sqrt{2}$

And foci are $(\pm ae, 0) \Rightarrow (\pm\sqrt{3}, 0)$

60.(AC) Let length of minor axis is ' $2b$ ' where b is parameter then equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and

$$e^2 = 1 - \frac{b^2}{a^2}$$

Now equation of tangents at extremities of latus rectum i.e. $(\pm ae, \pm \frac{b^2}{a})$ is

$$\frac{\pm xae}{a^2} + \frac{\pm yb^2}{ab^2} = 1$$

$$\Rightarrow \pm ex \pm y - a = 0 \Rightarrow (\pm y - a) \pm e(x - 0) = 0$$

Which represent a family of lines passing through fixed points $(0, \pm a)$

$$\Rightarrow \alpha = 0, \beta = \pm a$$

Again to find the locus of extremities of latus rectum, let $h = \pm ae, k = \pm \frac{b^2}{a}$

$$\Rightarrow h^2 = a^2 e^2, b^2 = \pm ak$$

$$\Rightarrow h^2 = a^2 - b^2, b^2 = \pm ak$$

$$\Rightarrow h^2 = a^2 \pm ak$$

Hence, locus of (h, k) is

$$x^2 = a^2 \pm ay \Rightarrow x^2 = a(a \pm y)$$

61.(AC) Let (h, k) be the point of intersection of two tangents then, equation of AB is

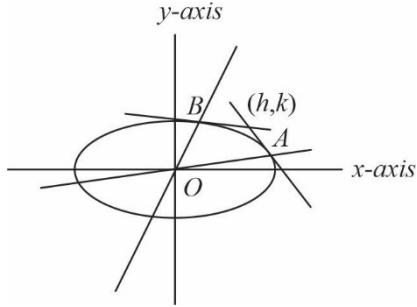
$$\frac{xh}{4} + \frac{yk}{1} = 1 \quad \dots\dots\dots (i)$$

Equation of ellipse is

$$\frac{x^2}{4} + y^2 = 1$$

Joint equation of OA and OB is (obtained by homogenizing equation of ellipse using (i))

$$\frac{x^2}{4} + \frac{y^2}{1} = \left(\frac{xh}{4} + \frac{yk}{1} \right)^2$$



$$\Rightarrow x^2 \left(\frac{h^2 - 4}{16} \right) + y^2 (k^2 - 1) + \frac{2hk}{4} xy = 0 \quad \dots\dots\dots (ii)$$

But combined equation of OA and OB is given i.e

$$x^2 + 4y^2 + \alpha xy = 0 \quad \dots\dots\dots (iii)$$

Since equations (ii) and (iii) represents same line

$$\therefore \frac{h^2 - 4}{16} = \frac{k^2 - 1}{4} = \frac{hk}{2\alpha}$$

$$\Rightarrow h^2 - 4 = 4(k^2 - 1) \Rightarrow \text{locus is } (x - 2y)(x + 2y) = 0$$

62.(AC) Let a common tangent be $y = mx + c$

$$\text{Then } C = \frac{a}{m} = a\sqrt{m^2 - 1}$$

$$\text{or, } m^4 + m^2 - 1 = 0$$

$$\text{or, } m = \pm \sqrt{\frac{1 + \sqrt{5}}{2}}$$

Thus there are always two common tangents irrespective of value of a .

63.(AC) Let P and Q be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ respectively so that the equation of the circle with PQ as its diameter is $(a - at_1^2)(x - at_2^2) + (y - 2at_1)(y - 2at_2) = 0$

It touches the directrix at T (where $x = -a$)

Hence $y^2 - 2ay(t_1 + t_2) + 4a^2t_1t_2 + a^2(1 + t_1^2)(1 + t_2^2) = 0$ has equal roots

$$\Rightarrow 4a^2(t_1 + t_2)^2 + 16a^2t_1t_2 - 4a^2(1 + t_1^2)(1 + t_2^2) = 0$$

$$\text{or } -2t_1t_2 - 1 - t_1^2t_2^2 = 0 \Rightarrow t_1^2t_2^2 + 2t_1t_2 + 1 = 0$$

$$\text{or } (t_1t_2 + 1)^2 = 0 \Rightarrow t_2 = -\frac{1}{t_1}$$

$$\Rightarrow y^1 - 2ay(t_1 + t_2) + a^2(t_1 + t_2)^2 = 0 \quad \text{i.e. } y = a\left(t_1 - \frac{1}{t_1}\right)$$

Hence the coordinates of P , T and Q respectively are :

$$(at_1^2, 2at_1), \left(-a, a\left(t_1 - \frac{1}{t_1}\right)\right), \left(\frac{a}{t_1^2}, -\frac{2a}{t_1}\right)$$

so that abscissa of P , T , Q are in G.P. and their ordinates are in A.P.

64.(ACD)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, b^2 = a^2(e^2 - 1)$$

$S(ae, 0)$ is vertex and $A(a, 0)$ is focus of the parabola

$$AS = (ae - a) = a(e - 1)$$

$$\therefore \text{Latus rectum of parabola} = 4a(e - 1)$$

$$\therefore \text{Equation of parabola is } y^2 = -4a(e - 1)(x - ae)$$

$$OS = ae$$

The parabola cuts positive y -axis at $y^2 = 4a^2e(e - 1)$

$$\therefore OK = \text{intercept on positive } y\text{-axis} = 2a\sqrt{e(e - 1)}$$

$$= \sqrt{4a(e - 1)ae} = \text{G.M of } OS \text{ and latus rectum.}$$

Section 4

Single Integer Answer Type

65.(2) Let (h, k) be the point.

Equation of chord of contact w.r.t. to ellipse $x^2 + 2y^2 = 1$ is given by

$$hx + 2ky = 1 \quad \dots\dots (1)$$

Making curve $x^2 + 2y^2 = 1$ homogenous with the help of (1) we can write

$$x^2 + 2y^2 = (hx + 2ky)^2$$

Now coefficient of $x^2 +$ coefficient of $y^2 = 0$

$$\Rightarrow (1 - h^2) + (2 - k^2) = 3$$

$$\Rightarrow h^2 + 4k^2 = 3$$

so locus of (h, k) is $x^2 + 4y^2 = 3$

66.(2) As circle touches normal so maximum radius = maximum distance of normal from origin = $|a - b|$

$$= |a^2 + 2a + 2 - a^2 - 1| = 5 \Rightarrow |2a + 1| = 5 \Rightarrow 2a + 1 = \pm 5 \Rightarrow a = 2, -3$$

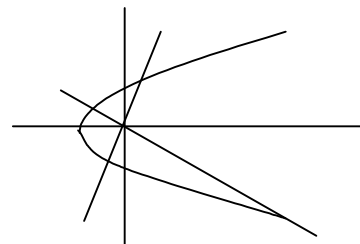
67.(4) Focus is origin any chord will be $y = mx$

length of this chord is

$$= \sqrt{1 + m^2} |x_2 - x_1| = \frac{\sqrt{1 + m^2} \cdot 4 \cdot \sqrt{1 + m^2}}{m^2} = \frac{4(1 + m^2)}{m^2}$$

Hence, $p = 4(1 + m^2)$

$$q = 4\left(1 + \frac{1}{m^2}\right) \Rightarrow (p - 4)(q - 4) = 16 \text{ or } pq = 4(p + q) \Rightarrow \frac{1}{p} + \frac{1}{q} = \frac{1}{4}$$



68.(5) Let $P \equiv (at_1^2, 2at_1)$; $Q \equiv (at_2^2, 2at_2)$

$$\text{Slope of tangent at } P = \frac{1}{t_1}$$

$$\text{Slope of } PQ = \frac{2}{t_1 + t_2}$$

$$\text{As per the given condition } \frac{1}{t_1} = -\frac{2}{t_1 + t_2}$$

$$\Rightarrow 3t_1 + t_2 = 0 \Rightarrow t_2 = -3t_1 \quad \dots\dots (1)$$

Let (h, k) be the middle point of PQ

$$\Rightarrow h = \frac{1}{2}a(t_1^2 + t_2^2) \text{ and } k = a(t_1 + t_2) \Rightarrow k = -2at_1 \text{ or } k^2 = 4a^2t_1^2$$

$$\text{and } 2h = a(t_1^2 + at_1^2) = 10at_1^2 \quad (\text{by (1)})$$

Eliminating ' t_1 ', we get

$$\frac{k^2}{2h} = \frac{4a}{10} \Rightarrow 5k^2 = 4ah \Rightarrow \text{locus is } 5y^2 = 4ax$$

69.(8) Let the middle point of chord be $(t^2, 2t)$

Mid point of chord must lie inside the ellipse

$$\Rightarrow t^4 + 8t^2 - 1 < 0$$

$$\Rightarrow t^2 \in (0, -4 + \sqrt{17}) \quad \dots\dots (1)$$

Also equation of the chord is $T = S_1$

$$\Rightarrow t^2x + 4ty = t^4 + 8t^2$$

Let this passes through $(\alpha, 0)$

$$\Rightarrow \alpha t^2 = t^4 + 8t^2 \Rightarrow t^4 + (8 - \alpha)t^2 = 0 \Rightarrow t^2 = 0 \text{ or } t^2 = \alpha - 8$$

$$\Rightarrow \alpha = t^2 + 8 \Rightarrow \alpha \in (8, 4 + \sqrt{17})$$

70.(7) Since $y^2 = 28x$ is the parabola, then equation of a focal chord AB which is inclined at an angle α with x -axis is

$$y - 0 = \tan \alpha (x - 7) \quad \dots\dots (1)$$

This satisfies $y^2 = 28x$

$$\Rightarrow (x - 7)^2 \tan^2 \alpha - 28x = 0$$

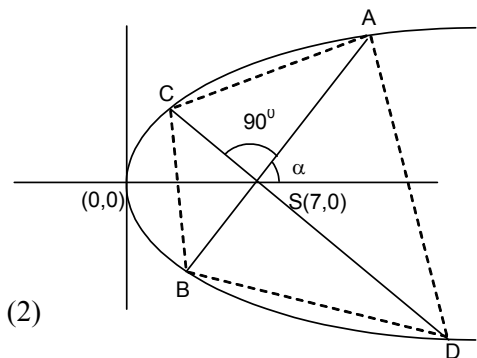
$$\Rightarrow x^2 \tan^2 \alpha - 14x(\tan^2 \alpha + 2) + 49 \tan^2 \alpha = 0 \quad \dots\dots (2)$$

If $A = (x_1, y_1)$ and $B = (x_2, y_2)$, then x_1, x_2 are the roots of equation (2)

$$\text{Now } x_1 + x_2 = \frac{14(\tan^2 \alpha + 2)}{\tan^2 \alpha} \text{ and } x_1 x_2 = 49$$

$$x_1 - x_2 = \sqrt{(x_1 + x_2)^2 - 4x_1 x_2} = \sqrt{\frac{14^2 (\tan^2 \alpha + 2)^2}{\tan^4 \alpha} - 49 \times 4}$$

$$x_1 - x_2 = 28 \cot \alpha \operatorname{cosec} \alpha$$



and $y_1 - y_2 = 28 \operatorname{cosec} \alpha$

$\Rightarrow AB = 28 \operatorname{cosec} \alpha$

Similarly, $CD = 28 \sec^2 \alpha$

Now area of $ACBD = \frac{1}{2} AB \cdot CD$

$$= \frac{1}{2} (28)^2 \operatorname{cosec} \alpha \cdot \sec^2 \alpha = \frac{784}{2} \operatorname{cosec}^2 \alpha \sec^2 \alpha = 1568 \operatorname{cosec}^2 2\alpha$$

Hence minimum value is 1568 and is attained when $\operatorname{cosec}^2 2\alpha = 1$ i.e. $\alpha = \frac{\pi}{4}$.

71.(6) Largest circle will be that circle which touches the ellipse at any point $(5 \cos \theta, 3 \sin \theta)$

$\Rightarrow$ Line joining points $(5 \cos \theta, 3 \sin \theta)$ and $(3, 0)$ will be perpendicular to tangent of ellipse at point $(5 \cos \theta, 3 \sin \theta)$

so that $\frac{3 \sin \theta}{5 \cos \theta - 3} \times \frac{-3 \cos \theta}{5 \sin \theta} = -1$ or $\cos \theta = \frac{15}{16}$ and $\sin \theta = \frac{1}{4}$

so that point on ellipse is $\left(\frac{75}{16}, \frac{3\sqrt{31}}{16} \right)$

Hence radius of circle is $\frac{3\sqrt{7}}{4}$

72.(2) Let the given circle be the director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{9-a^2} = 1$ (1)

If $(3 \cos \theta, 3 \sin \theta)$ is a point on the given circle, then its chord of contact w.r.t. ellipse (1) is

$$\frac{3x \cos \theta}{a^2} + \frac{3y \sin \theta}{9-a^2} = 1 \text{ or } \frac{2x \cos \theta}{\frac{2a^2}{3}} + \frac{y \sin \theta}{\frac{9-a^2}{3}} = 1.$$

This line is a tangent to the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$

$\Rightarrow \frac{2a^2}{3} = 4$ and $\frac{9-a^2}{3} = 1 \Rightarrow a^2 = 6$

Hence the equation of the ellipse (1) is $\frac{x^2}{6} + \frac{y^2}{3} = 1$

73.(3) Given a hyperbola $xy = c^2$

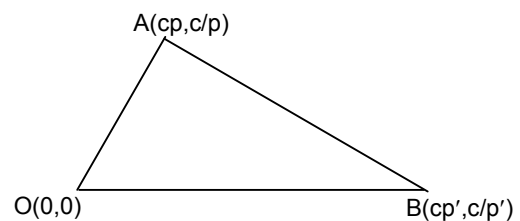
let a point $A(cp, c/p)$ be on hyperbola $xy = c^2$
normal at point A , again cut the hyperbola at

point $B(cp', c/p')$ where $p' = \frac{-1}{p^3}$

slope of line $OA = \frac{1}{p^2}$ and slope of line $OB = p^6$

$\Rightarrow \tan(\angle AOB) = \frac{1/p^2 - p^6}{1 + p^4} = \frac{1-p^8}{p^2}$ (1)

Now slope of $AB = p^2$



$$\Rightarrow \tan(\angle OAB) = \frac{p^2 - 1/p^2}{1 + 1} = \frac{p^4 - 1}{2p^2} \quad \dots\dots\dots (2)$$

$$\Rightarrow \frac{\tan(\angle AOB)}{\tan(\angle OAB)} = -2$$

Now, given $\angle AOB = \alpha$ ($\because$ normal chord subtend an angle α to the origin)

$$\Rightarrow \frac{\tan \alpha}{\tan A} = -2 \Rightarrow \frac{\sin(\alpha + A) + \sin(\alpha - A)}{\sin(\alpha + A) - \sin(\alpha - A)} = -2 \Rightarrow \frac{\sin(\alpha - A)}{\sin(\alpha + A)} = 3$$

74.(0) Let hyperbola $xy = -c^2$ and parabola $x^2 = 4by$ cuts at a point (x_1, y_1)

$$\text{so that } x_1 y_1 = -c^2 \text{ and } x_1^2 = 4by_1 \quad \dots\dots\dots (1)$$

and slope of tangents to the hyperbola and parabola at the point (x_1, y_1) are respectively

$$\tan \theta_1 = \frac{-y_1}{x_1} \text{ and } \tan \theta_2 = \frac{x_1}{2b}$$

$$\text{From } x_1^2 = 4by_1, \text{ we get } \tan \theta_1 = \frac{-x_1}{4b}$$

$$\text{So that } 2 \tan \theta_1 + \tan \theta_2 = -\frac{x_1}{2b} + \frac{x_1}{2b} = 0$$